

COURSE OUTCOMES

After completion of course, students will be able to:

1. Analyse various types of data, and its preprocessing.
2. Illustrate and analyse various applications of data mining.
3. Analyse and compare various classification models in data mining.
4. Understand developments in big data technologies.
5. Familiarize the concepts of machine learning using R/Python.
6. Analyse and make use of deep learning using R/Python.
7. Familiarize concepts of social network analysis and sentiment analysis

Module-1

Data Mining-Purpose-Various phases of data mining - supervised vs. unsupervised - learning - Data Warehouses - OLAP - Multidimensional databases - Data Pre-processing-Case studies in data pre-processing-Dimensionality reduction techniques-Chisquare, PCA, LDA. Different applications of data mining..Introduction to machine learning concepts of pattern recognition,.

Module-2

Association Rules mining-Apriori algorithm-Examples-Improvements for apriori algorithm- Classification concepts, mathematical notions and case studies-Decision trees, Neural networks, Naïve Bayes classifier, KNN, Support vector machines, Maximum likelihood classifier

Module-3

Cluster Analysis-K-Means algorithm-Example and suggestions for improvements- A Categorization of Major Clustering Methods, Partitioning Methods, Hierarchical Methods, Density-Based Methods (DBSCAN), Time series mining, Graph Mining, Reinforcement learning, Hidden markov models- Case studies-Introduction to rough sets-Mathematical Notions.

Module-4

Introduction to cloud computing-Services from a cloud- Big Data-definition-data bases for the big data platform-Introduction to Hadoop its architecture and ecosystem. MapReduce-basic concepts- Introduction to Spark-Deep learning-Concepts-CNN and RNN- Typical use cases. Introduction to social network analysis, sentiment analysis

References:

1. Jiawei Han and Micheline Kamber, Data Mining: Concepts and Techniques – 3rd Edition, Elsevier, 2011. ISBN: 9780123814807
2. Ian H.Witten et.al, Data Mining: Practical Machine Learning Tools and Techniques, Morgan Kofman publishers. 4th edition, 2016. ISBN: 978-0123748560
3. Ian Good fellow et.al, Deep learning, First edition, MIT Press, 2016. ISBN:978-0262035613
4. Tom White, Hadoop: The definitive guide, 3rd edition, OReilly Publishers, 2012. ISBN : 978-1449311520
5. Karau. H. et.al, Learning Spark: lightning-fast big data analysis, OReilly Publishers, 2015 hadoop, tom white. ISBN: 978-9351109945